6 Automated and emerging technologies

In this chapter, you will learn about:

- ★ automated systems
  - the use of sensors, microprocessors and actuators in automated systems
  - the advantages and disadvantages of using automated systems in given scenarios
- ★ robotics
  - what is meant by robotics
  - the characteristics of a robot
  - the roles, advantages and disadvantages of robots
- ★ artificial intelligence
  - what is meant by artificial intelligence (AI)
  - the main characteristics of AI
  - the basic operation and components of AI systems to simulate intelligent behaviour.

Whether it's robots welding cars in a factory, autonomous grass cutters or smart road signs, the effects of automated systems can be seen all around us. These automated systems are becoming increasingly sophisticated and complex. This chapter will consider examples of automated systems, robotics and artificial intelligence (AI) and how they affect our everyday lives. This is by no means an exhaustive list and the reader is advised to try to keep up to date with all the latest developments.

6.1 Automated systems

6.1.1 Sensors, microprocessors and actuators

An **automated system** is a combination of software and hardware (for example, sensors, microprocessors and actuators) that is designed and programmed to work automatically without the need of any human intervention. However, such systems often involve human monitoring.

Link

See Section 3.2 for more details on sensors, microprocessors and actuators.

The role of sensors, microprocessors and actuators was discussed at great length in Section 3.2. It may be worth the reader revisiting this part of the book before continuing with this chapter; you should remember that:

- Sensors are input devices that take readings from their surroundings and send this data to a microprocessor or computer. If the data is analogue, it is first converted into a digital format by an analogue-digital converter (ADC).
- The microprocessor will process the data and take the necessary action based on programming.
- This will involve some form of output, usually involving signals sent to actuators to control motors, wheels, solenoids, and so on.

Advice

- On first sight, all of the examples in 6.1 will appear very complicated.
- However, you will not learn any of the industrial or scientific processes described fully in this chapter. Any processes used in any questions will be fully described to you (possibly including a diagram).
- You basically need to go through each example carefully and understand the processes taking place. In other words, what is the interaction between the sensors, actuators and microprocessor/computer to allow the process to take place.
- On completion of Section 6.1 the important learning process is to understand the sensor, actuator and microprocessor interaction; to this end, you need to do two things:
  - for each example, complete a table as follows:

Example	Which sensors are used?	What is the function of the actuators?	What is the function of the computer?	Additional notes

  - then try the activities 6.1 to 6.7 to make sure you fully understand the processes going on.

  You will then find out that the examples were much easier than they appeared at first.

6.1.2 Advantages and disadvantages of automated systems

In this section, a number of examples will be used to show the advantages and disadvantages of using automated systems. This list is by no means exhaustive, and simply intends to show the role of sensors, microprocessors (or computers) and actuators in the following application areas:

- industrial
- transport
- agriculture
- weather
- gaming
- lighting
- science.

Industrial applications

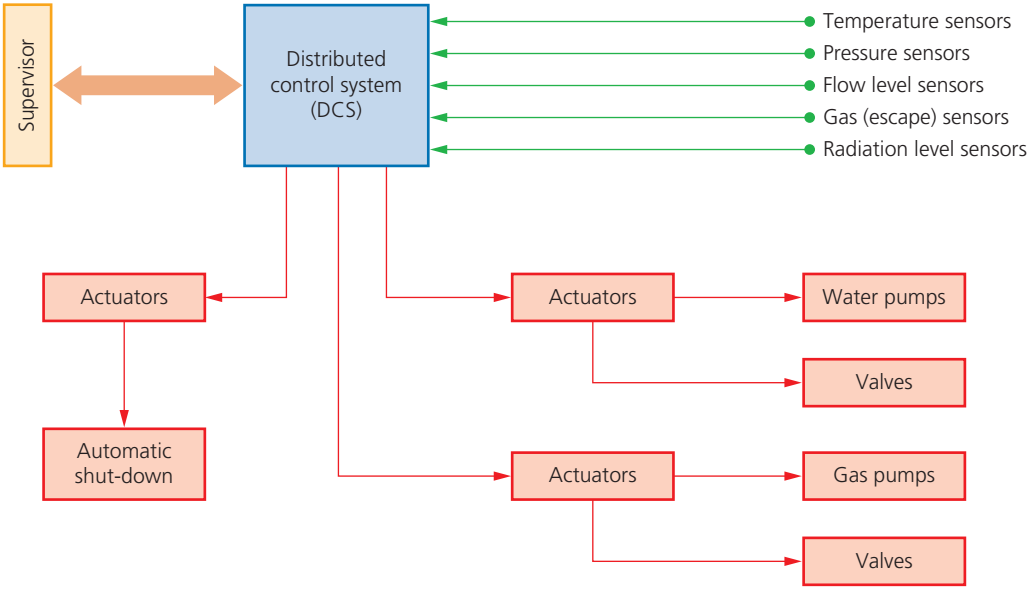
Automated systems are used in a number of industrial applications. Many of the automated systems involve robotics, which is covered in more depth in Section 6.2.

In recent years, the focus on increased automation has led to improved quality and flexibility. For example, in the manufacture of car engines, when done manually, the installation of pistons into the engine had an error-rate of ~1.5%; with automated systems, the error-rate has fallen to 0.00001%.

We will now consider two very different industrial applications.

Example 1: A nuclear power station

A key use of automated systems is in the control and monitoring of a nuclear power station. This is a good example, since automation gives increased safety both in the process itself and to the workforce. At the centre of the system is a **distributed control system (DCS)**. DCS is essentially a powerful computer that has been programmed to monitor and control the whole process with no human interaction required:



▲ **Figure 6.1** Nuclear power station (automated system)

Data from a number of sensors is sent to a DCS (computer) – if the data is analogue, it must first be converted into digital format using an ADC. The DCS will have access to a large database containing operational data and parameters. If any action needs to be taken, then signals will be sent to the appropriate actuators to operate pumps, valves or even an emergency shutdown system. The key here is that the system is fully automated. A human operator (the supervisor) will sit in a remote control room where a schematic of the process will show on a large screen. While the process is fully automatic, the supervisor can still override the DCS and shut down the process.

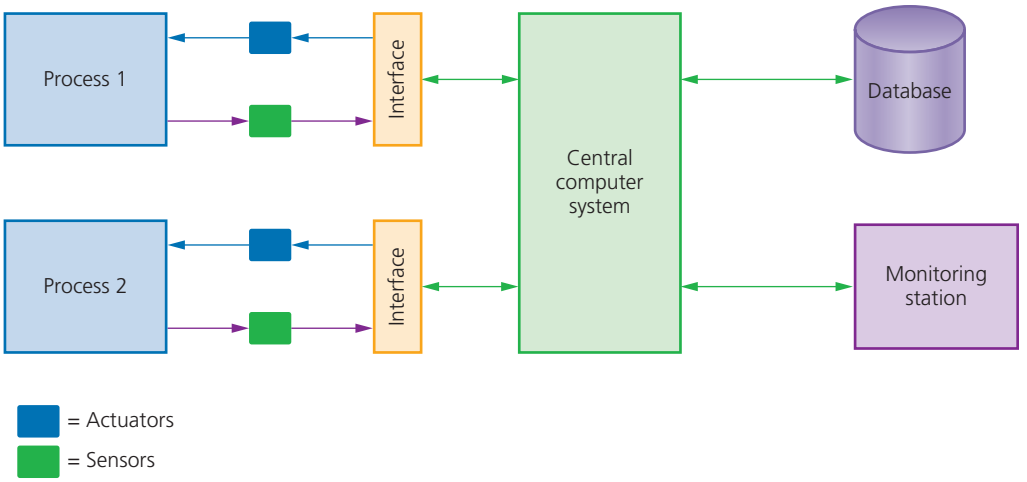
The main advantages of this automated system are:

- much faster than a human operator to take any necessary action
- much safer (an automated system is more likely to make timely interventions than a human; it also keeps humans away from a dangerous environment)
- the process is more likely to run under optimum conditions since any small changes needed can be identified very quickly and action taken
- in the long run, it is less expensive (an automatic system replaces most of the workforce who would need to monitor the process 24 hours a day).

The main disadvantages of this automated system are:

- expensive to set up in the first place and needs considerable testing
- always possible for a set of conditions to occur that were never considered during testing which could have safety implications (hence the need for a supervisor)
- any computerised system is subject to cyberattacks no matter how good the system (one way round this is to have no external links to the DCS; although the weak link could potentially be the connection to the supervisor)
- automated systems always need enhanced maintenance which can be expensive.

Example 2: Manufacture of paracetamol



▲ **Figure 6.2** Manufacture of paracetamol (automated system)

This automated system also depends on sensors, a computer, actuators and software. Process 1 is the manufacture of the paracetamol. Process 2 is the making of the solid tablets. Both processes are monitored by a number of sensors that send their data back to a central computer. The computer consults its database to ensure both processes are operating within correct parameters. Any necessary action is taken by the computer, sending signals to the appropriate actuator to operate pumps, valves, heaters, stirrers or pistons to ensure both processes can operate without any human intervention. Again, this system uses a remote monitoring station manned by an operator. The system is fully automated, but the operator can override the central computer system if necessary.

The main advantages of this automated system are:

- much faster than a human operator to take any necessary action
- much safer (an automated system is more likely to make timely interventions than a human if necessary; it also keeps humans away from a potentially dangerous environment)
- the process is more likely to run under optimum conditions since any small changes needed can be identified very quickly and action taken
- in the long run, it is less expensive (an automatic system replaces most of the workforce who would need to monitor the process 24 hours a day)
- more efficient use of materials
- higher productivity
- more consistent results.

The main disadvantages of this automated system are:

- expensive to set up in the first place and needs considerable testing
- always possible for a set of conditions to occur that were never considered during testing which could have safety implications (hence the need for a monitoring station)
- automated systems always need enhanced maintenance which can be expensive
- any computerised system is subject to cyberattacks no matter how good the system.

There are many other examples; the above two examples can be applied to many other industrial processes.

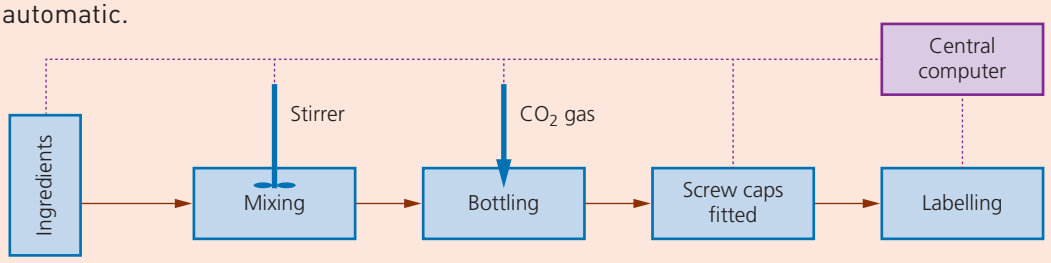
Activity 6.1

A company manufactures fizzy drinks that are then labelled and bottled:

The system is fully automatic. Sensors are used to ensure the correct amount of each ingredient is added. A stirrer is activated whenever ingredients are added. The bottling plant is again monitored by sensors to ensure the correct amount of drink is added to each bottle, and that the right amount of carbon dioxide gas is added to each bottle. The whole process is computer-controlled and is totally automatic.

a Describe how the sensors, actuators and central computer would be used to monitor and control this bottling plant automatically. You may wish to add/show sensors and actuators in the diagram given above.

b Finally, describe the advantages and disadvantages of fully automating this bottling plant (you can assume that none of the ingredients used in making the drink are harmful).



▲ **Figure 6.3**

Find out more

Find out:

i which sensors are used in modern aircraft when using auto pilot

ii how sensors and actuators are used to control the flaps, throttle and rudder

iii why airplanes use the 'three computer system'.

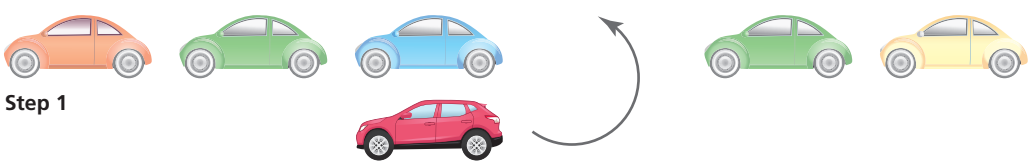
Transport

As with industrial processes, many of the automated systems in transport refer to robotic systems (for example, autonomous buses/cars, autonomous trains and unpiloted aircraft). These will be considered in Section 6.2.

But automated systems are still used in manually controlled transport, which includes cars, buses/lorries, trains and aircraft. (Examples 3 and 4 which follow, will use cars as the application.)

For example, modern trains will use an automatic signal control system. If the driver of the train goes through a red (stop) light, then the computer will automatically stop the train. This will make use of sensors at the side of the track sending signals to the on-board computer; actuators will be used to apply the brakes. Airplanes extensively use automatic pilots, which control the wing flaps, throttle and rudder to maintain the correct height, speed and direction.

? Example 3: Self-parking cars



Step 1



Step 2

▲ **Figure 6.4** Self-parking cars

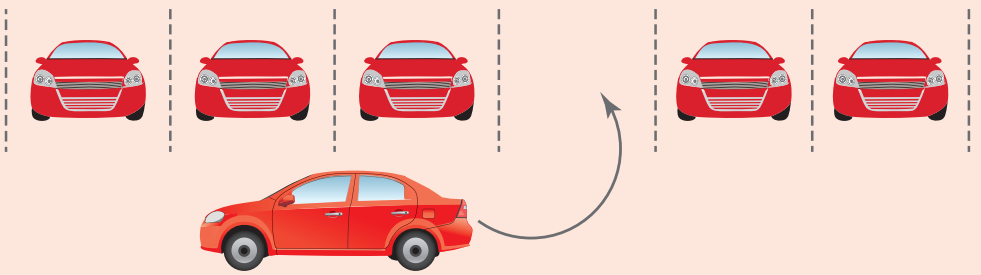
The driver goes along the row of parked cars. On-board sensors and cameras gauge the size of any parking spaces, and the on-board computer warns the driver if a suitable space has been found. The driver then selects auto-parking and the on-board computer takes over. Actuators are used to operate the steering rack, brakes and throttle under the full control of the computer. This allows the car in Figure 6.4 to go from Step 1 to Step 2 automatically and complete the parking manoeuvre with no driver intervention.

Sensors in the bumpers of the car are both transmitters and receivers. The sensors transmit signals that bounce off objects and are reflected back. The car's on-board computer uses the amount of time it takes for the signal to return to the sensor to calculate the position of any objects. The sensors give the computer a 3D image of its surroundings. This allows the car to fit into its parking space automatically with no driver intervention. (Note: cheaper and older self-parking systems are not fully automatic; they require the driver to operate the brakes and throttle manually and only control the steering.)

The main advantages of this automated system are:
- allows the same number of cars to use fewer parking spaces
- avoids traffic disruption in cities (a manually controlled car takes several seconds to fit into a parking space)
- cars can fit into smaller spaces
- fewer dents and scratches to cars (reduced insurance claims)
- safer system since sensors monitor all objects, including young children (the car's manoeuvre will be stopped if any new object is encountered)
- very consistent results.

The main disadvantages of this automated system are:
- over-reliance on automated systems by the driver (loss of skills)
- faulty/dirty sensors or cameras can send false data/images to the on-board computer which could lead to a malfunction
- kerbing of wheels is a common problem since the sensors may not pick-up low kerbs
- expensive option that doesn't really save the driver any money
- requires additional maintenance to ensure it functions correctly at all times.

Activity 6.2



▲ **Figure 6.5**

This time the driver needs to reverse in between two other cars and park at 90° to the direction of travel.

Describe what additional information is needed to allow the car to park between two cars within the parallel lines.

What extra sensor device(s) might be needed to give this additional information?

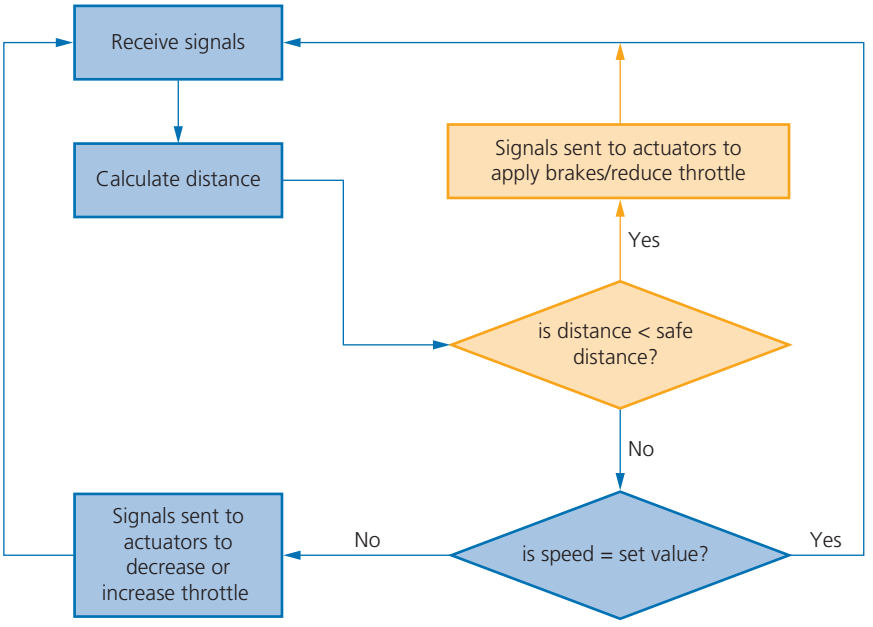
Example 4: Adaptive cruise control



▲ **Figure 6.6** Adaptive cruise control

Adaptive cruise control makes use of sensors, an on-board computer and actuators to allow a car to remain a safe distance from another vehicle.

The driver will set a cruising speed (for example, 100 kph) on his touch screen in the car. Lasers (set into the bumpers of the car) are used to send out signals constantly. The lasers bounce off the vehicle in front of the car and are reflected back to the car's sensors. The time taken for the signal to bounce back is used by the on-board computer to calculate the distance between the two vehicles. If the car is getting too close to the vehicle in front, the computer will send signals to slow the car down. This is done by actuators applying the brakes and/or reducing the throttle. If the distance between vehicles is greater than the safe distance, the computer will check to see if the current speed equals the value set by the driver. If the speed is different to the set speed, the computer sends signals to the actuators to increase or decrease the throttle.



▲ **Figure 6.7** Flowchart showing how adaptive cruise control works

Activity 6.3

Assuming the following stopping distances:

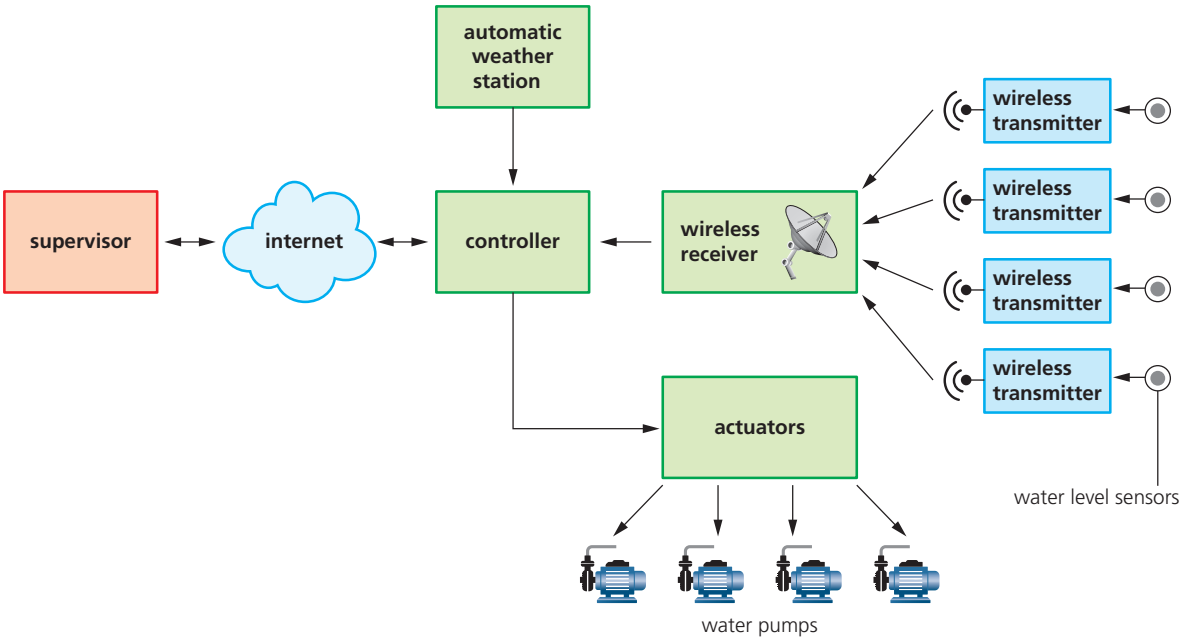
Car's speed (kph)	Stopping distance (metres)
40	17
50	24
60	32
70	42
80	53
90	64
100	77
110	92
120	107

Using figure 6.7 and the data in the table above, explain how the on-board computer would calculate if a car was too close to the vehicle in front and apply the brakes if necessary. (It is not necessary to show all the maths, it is sufficient to explain what would need to be done. However, this could be a class exercise using a spreadsheet or small computer program).

Agriculture

There are many examples of the use of automated systems in agriculture. Again, many of the systems involve robotics, which is fully described in Section 6.2. We will now consider one important example that is being used in Brazil to irrigate crops automatically.

Example 5: Automated system used in the Brazil irrigation system



▲ **Figure 6.8** Automatic irrigation system

The watering of crops (irrigation) is fully automatic and also involves considerable amounts of wireless transmission. This allows the system to be used in very remote areas that are vast in size – some of the fields are more than 10 km^2 in area.

Data from an automatic weather station (see next example) is received by the **controller** (a computer system) every ten minutes. This is particularly important if very wet or very dry conditions are being predicted or detected by the weather station. Ultrasonic water level sensors are used in the crop fields that measure the amount of water in the irrigation channels. The sensors send their data back via wireless transmitters. This data is then picked up by the wireless receiver, which sends the data back to the controller. The controller then uses this data, together with the data from the weather station, to decide whether it is necessary to stop or start a series of water pumps. This is done by sending signals to actuators, which operate the pumps. Although the whole system is fully automatic, a supervisor still monitors the process remotely. Using a schematic of a number of processes on a computer screen and via internet links to the controllers, the supervisor can oversee several irrigation processes from one central point. If the supervisor wishes to further increase or reduce the water supply in any of the irrigation systems, they can override the controller if necessary.

The main advantages of this automated system are:

- reduced labour costs since the system only needs a supervisor to monitor vast areas (if any maintenance is needed, then a dedicated team can cover all of the irrigation systems rather than having a separate team for each system)
- better and more efficient control of the irrigation process
- better control of precious resources, such as water

- faster response than a human having to manually check many kilometres of irrigation channels
- safer (temperatures in the fields could be 40°C and other risks could exist)
- different crops may require different irrigation requirements (for example, rice crops need flooding conditions, whereas orange trees like dry conditions); it is possible to program the controllers so that different growing conditions can be maintained simultaneously.

The main disadvantages of this automated system are:

- expensive to set up initially (expensive equipment needs to be bought)
- very high maintenance costs are associated with automated systems (also require specialist technicians if a fault occurs, which could be a problem in some remote areas of the world)
- increased need to maintain the water channels to ensure the system works correctly at all times (a blocked or collapsed channel wouldn't be picked up by the automated system, which could result in some areas being over-watered and some areas being starved of water).

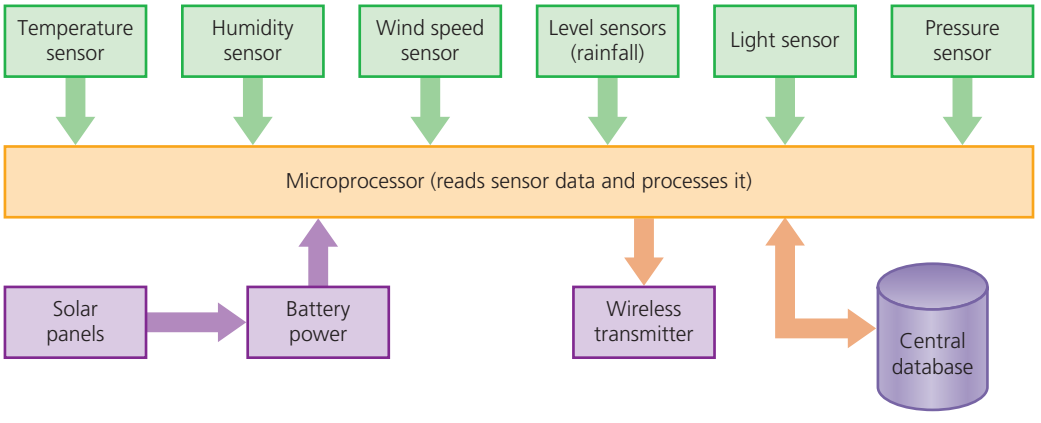
Weather (stations)

Automated weather stations are designed to save labour and to gather information from remote regions or where constant weather data is a requirement. Automated weather stations require a microprocessor, storage (database), battery (usually with solar-powered charging) and a range of sensors:

- thermometer (to measure temperature)
- anemometer (to measure wind speed)
- hygrometer (to measure humidity)
- barometer (to measure air pressure)
- level sensor (to measure rain fall)
- light sensor (to measure hours of daylight).

The data from sensors is all sent to a microprocessor; any calculations are then done (for example, calculate hours of daylight, actual rainfall and wind direction). The data from the sensors and the calculated values are then stored on a central database. Some automated weather stations are sited near airports, where reports are sent out automatically every five minutes to pilots in the vicinity of the airport.

The only part of the weather station that needs to use actuators is the 'tipping bucket rain gauge'. At a pre-determined time interval, a signal is sent from the microprocessor to an actuator to operate a piston, which tips a bucket that was collecting rain water. The water is tipped into a vessel where level sensors are then used to measure the amount of rainfall that fell during the required time interval.



▲ **Figure 6.9** Automated weather station

Activity 6.4

Find out all of the advantages and disadvantages of using automated weather stations.

Gather all the results from the whole class, and put your results into a table as follows:

Advantages	Disadvantages

Activity 6.5

1 Describe how automated weather stations can be used in the fight against climate change.

2 A large greenhouse is being used to grow tomatoes under controlled conditions.

For optimum growth, the tomatoes require the right lighting levels, correct temperature and regular watering. Describe how automated systems could be used to ensure the correct growing conditions are maintained. The only human involvement would be as a remote supervisor.

(You might find it useful to draw a diagram of your automated system showing sensors, microprocessor and any actuators.)

3 A new car being developed utilises many automated systems. In particular:

i the ability to recognise road signs

ii the ability to predict when to change to the correct gear for the road condition.

Describe how sensors, actuators and an on-board computer allow the car to take the necessary action automatically if:

- it encounters road works with a speed limit of 60 kph so that the car maintains the correct speed
- the car is on a very twisty road where the correct gear is needed for optimum performance.

4 One example of an automated system is the control of entry and exit to a car park. Cameras take a photograph of a car's number plate on entry before opening a barrier. At the exit, another camera captures a car's number plate before raising the barrier. Describe how sensors, cameras, actuators and a microprocessor can be used to:

i control the raising and lowering of the entry and exit barriers

ii ensure that the car exiting the car park has paid the correct parking fee before it can exit

iii describe the advantages and disadvantages of this car parking system.

Gaming

Gaming devices involve sensors to give a degree of realism to games:

- **accelerometers** (these measure acceleration and deceleration and therefore measure and respond to tilting the gaming device forward/backward and side to side)
- proximity sensors (used in smart touch pads; here electrodes are embedded in touch pads that can detect hand/finger position thus increasing user awareness).

Embedded accelerometers and proximity sensors (together with a microcontroller) in games consoles allow increased human interaction with the game. This allows players to take actions that simulate real events happening, giving a more immersive games experience.

Activity 6.6

What are the advantages and disadvantages of using these immersive games consoles?

Are any other sensors used in games consoles other than accelerometers and proximity sensors?

Lighting

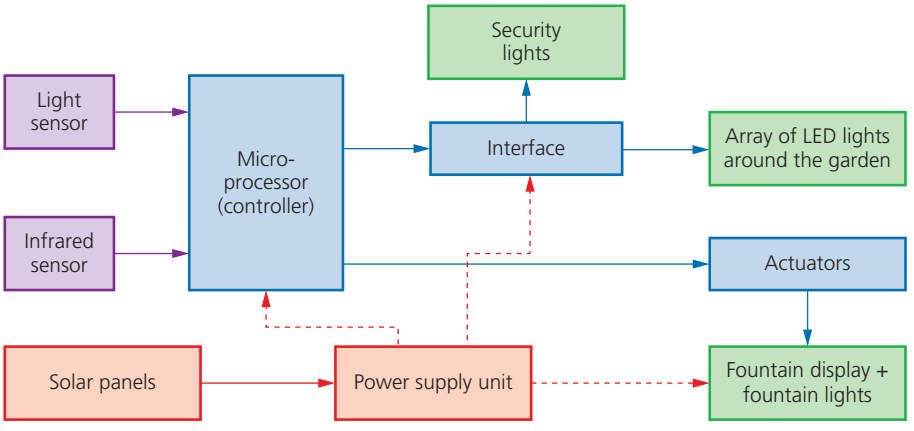
Microprocessor-controlled lighting was discussed in Section 3.1.5 (Embedded systems). The example used in Chapter 3 was the control of lighting in an office using:

- light sensors (to automatically switch lights on or off depending on the ambient lighting)
- motion sensors (to automatically turn lights on in a room when somebody enters)
- infrared sensors (to be used either as a motion detector or as part of the security system).

? Example 6: Lighting system in a house

The example we will consider here is used in a house:

- where lights in the garden are turned on automatically when someone enters the garden or it turns dark
- where a lighting show is part of a microprocessor-controlled water fountain display; the lighting only comes on when it becomes dark.



▲ **Figure 6.10** Automatic lighting system

As it becomes dark, the light sensor value will change, and the microprocessor will send signals to the interface to control the array of LED lights around the garden. Data from the infrared sensor would also be used (during day and night) as a security device whenever the house is unoccupied.

Also, as it becomes dark, the lighting show in the fountain could also be initiated. This could involve a pre-programmed display involving changing colours under the control of the microprocessor. The fountain display itself will also be under microprocessor control with signals being sent to actuators to turn water pumps on and off according to the installed program. The whole system will be fully automated.

The main advantages of this automated system are:

- it is possible to control light sources automatically
- a reduced energy consumption (since lights are only turned on when necessary)
- wireless connections can be chosen which are much safer (no trailing wires)
- longer bulb life (due to dimming or switching off when not in use)
- possible to program new light displays for various occasions.

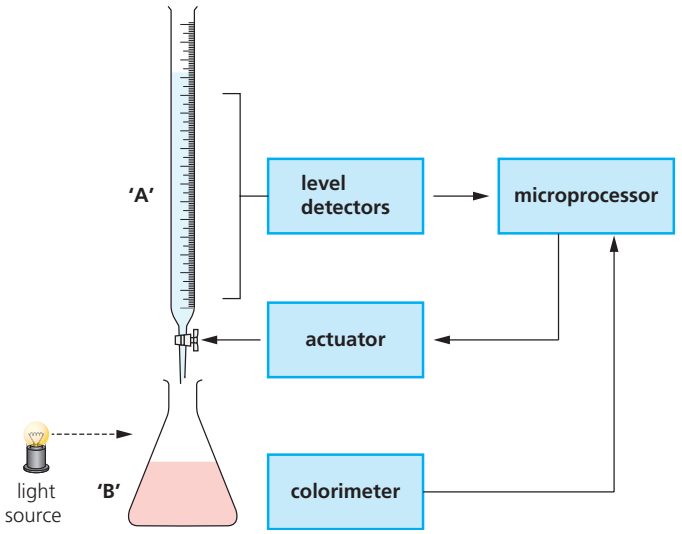
The main disadvantages of this automated system are:
- expensive to set the system up in the first place
- if wireless connections chosen (for safety reasons), they can be less reliable than wired systems
- to ensure consistent performance, the automated system will require more maintenance (which can be expensive).

Science

Automated systems in scientific research are widely used. There are literally thousands of possible applications. The example we will use here is the automatic control of a laboratory experiment which requires accuracy and repeatability.

Example 7: Chemical process in a laboratory

Imagine an experiment in a pharmaceutical laboratory where two chemicals are reacted together in a vessel. One of the chemicals is being added from a piece of equipment ('A') known as a **burette** (which has a tap to control the flow of liquid; the tap is operated automatically using a small actuator) to a reaction vessel ('B'). Once the reaction is complete, it turns a bright orange colour (see Figure 6.11). The whole process is under microprocessor control:



▲ **Figure 6.11** Pharmaceutical laboratory experiment

The level sensors measure how much liquid is being added from 'A'; this data is sent to a microprocessor. Readings are also sent to the microprocessor from a colorimeter next to vessel 'B' (this instrument checks the colour of the chemical produced). The microprocessor controls the opening and closing of the tap in 'A'; this is done by sending signals to an actuator that operates the tap. This means the microprocessor has automatic control of the experiment with no human interaction needed. This type of automated system is of great benefit to pharmaceutical companies when developing new drugs and vaccines (several experiments can be carried out at the same time with only one person needed to monitor the system).

The main advantages of this automated system are:
- more consistent (repeatable) results
- less dangerous (especially if the chemicals used are hazardous)
- faster results (several different experiments can be done simultaneously)
- automatic analysis of the results is possible
- fewer highly trained staff needed for each experiment
- results/experiments can be monitored anywhere in the world in real time.

The main disadvantages of this automated system are:
- less flexible than when using human technicians
- security risks are always present if the data is being shared globally
- equipment can be expensive to buy and set up in the first place.

Finally, there are many automated systems being used in both industry and scientific research that incorporate artificial intelligence (AI). It is therefore worth considering the generic advantages of using AI in these automated systems (also refer to Section 6.3):

- ability to access and store vast amounts of facts (very important in research)
- they are able to learn from huge amounts of available data that would overwhelm humans (or at the very least take them many months/years to do the same analysis)
- they are able to see patterns in results that could be missed by humans.

While all of this is positive, there are a few disadvantages in this approach:

- a change in skills set (is it the human or the AI that controls the research?)
- AI is dependent on the data which trains it.

Activity 6.7

1 a Name suitable sensors for each of the following automated systems.

i Manufacture of a new vaccine that requires the mixing of four liquids in the ratio 1:2:3:4 as a single batch. The four liquids must be totally mixed and the temperature must be maintained at 35°C (+/- 1°C) which is a critical temperature.

ii A lighting display has been set up in one room of an art gallery. A random sequence of different coloured lights is under microprocessor control. The display in the room only switches on when visitors walk into the room; at the same time, the room lights are also dimmed to give the most dramatic effect of the light display.

iii A train uses automatic twin-doors. Both doors open automatically when the train stops. Both doors close again when no-one is still boarding or leaving the train. The doors have a safety mechanism so that a passenger cannot become trapped between the two closing doors. The train can only move off when every door on the train has been safely closed.

b For **each** application in **part a**, give **two** advantages and **two** disadvantages of using automated systems.

2 The eight statements on the left-hand side of the table are either true or false. Tick (✓) the appropriate box to indicate which statements are true and which statements are false.

Statement	True	False
automated systems lead to less consistent results or less consistent products		
automated systems are more expensive to set up than traditional manual systems		
automated systems would be quickly overwhelmed by the amount of data presented to it		
automated systems are inherently less safe than manual systems		
automated systems generally require enhanced maintenance when compared to manual systems		
automated systems allow processes to run at optimum conditions at all times		
software failures, due to unforeseen conditions, are unlikely to impact on an automated system		
automated systems will react more quickly to unusual process conditions		